

Final Individual Report

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Lewis University

CPSC 49300 – Computer Infrastructure Capstone

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Final Individual Report

This report summarizes my individual experience and contributions to the Spring 2025 Computer Infrastructure Capstone project. It includes a description of the project's goals, an overview of each team member's role, and a personal reflection on the technical, collaborative, and service-learning aspects of the experience.

1. Project Description

The Spring 2025 Computer Infrastructure Capstone project focused on designing, configuring, and documenting a secure, scalable network infrastructure for a simulated non-profit tutoring organization operating three centers in Romeoville, Chicago, and Aurora, Illinois. The initiative combined theoretical research in network architecture, topologies, and cybersecurity protocols with a real-world service-learning component that emphasized client-focused design and regulatory compliance.

Our primary objective was to deliver a cost-effective infrastructure capable of supporting staff, tutors, and students at all three locations, each with its own VLAN to segment traffic and protect sensitive data. The network was designed to meet FERPA, SOPPA, and Illinois PIPA compliance standards and included a DMZ web server, centralized storage server, enterprise firewalls, site-to-site VPN connectivity, wireless access, and endpoint security. Additional requirements included surveillance systems, support for non-cash payment processing, and secure remote access.

To achieve these goals within a \$60,000 budget (with an optional \$20,000 expansion), we conducted site evaluations, selected appropriate hardware and software, and developed detailed

technical documentation. This included physical and logical topologies, subnetting schemes, security controls, and implementation timelines. The Romeoville center, due to its size and infrastructure, was selected as the central hub.

2. Team Contributions

Team Members

- Boris Cazarin
- Chelsea Barry
- Daniel Monbrod (myself)

Boris Cazarin

- Led physical network design and implementation planning.
- Developed cable run maps, switch and server layouts for Romeoville and Chicago.
- Contributed to documentation on physical security and firewall selection.
- Assisted with wireless heatmap testing using NetSpot and validation with Wireshark.

Chelsea Barry

- Led theoretical research on OSI/TCP/IP models, LAN/WAN technologies, and VPN protocols.
- Authored the literature review and user-facing documentation.

- Focused on privacy law compliance (FERPA, SOPPA, Illinois PIPA).
- Supported final formatting and assisted in network diagram creation.

Daniel Monbrod (My Role)

- Designed and configured the overall security architecture including Fortinet firewalls, integrated IDS/IPS, DLP, and web filtering.
- Led the implementation of IPsec VPN tunnels and VLAN segmentation across all sites.
- Created the IP addressing and subnetting scheme for efficient routing and access control.
- Configured Microsoft Sentinel for centralized SIEM and documented alerting procedures.
- Integrated user authentication policies using FortiGate captive portals and access control lists.
- Authored and presented key sections of the final report and presentation, particularly on network security, SIEM, and firewall strategy.

During the semester, I experienced a short-term medical absence that briefly impacted my ability to contribute. Upon recovery, I reviewed all missed progress, quickly re-synced with my team, and worked diligently to reinforce our security implementation, documentation, and testing. This period tested my resilience but also reaffirmed the importance of communication and accountability in collaborative work.

3. Reflection

a. The Project Itself

This project offered a highly realistic and technical challenge. Designing an end-to-end infrastructure that included network segmentation, secure remote access, regulatory compliance, and documentation forced us to think like IT consultants. It pushed me to apply principles from multiple prior courses—from subnetting and logical topology design to SIEM integration and data protection strategies. I leave this experience with a stronger technical foundation and greater confidence in my ability to architect real-world network environments.

b. Dealing with Clients

Although our client was represented by the instructor, the process still required clear, professional communication. We were responsible for interpreting vague or non-technical input and translating it into actionable technical solutions. This helped refine my ability to present ideas concisely, clarify expectations, and balance the client's needs with technical realities and constraints.

c. Providing Community-Based Learning

Knowing that our infrastructure served an educational mission made the work especially meaningful. Our design supported tutoring centers intended to enhance learning outcomes for K–12 and college students—especially in underserved areas. This gave weight to decisions we made about reliability, accessibility, and cost-effectiveness, and reminded me that technology can and should serve human needs.

d. Working with a Team

This was one of the most cohesive and supportive teams I've worked with. Each member took responsibility for their role, but we also routinely collaborated, reviewed each other's work, and filled in gaps where needed. After returning from my brief absence, the team welcomed me back without hesitation, which motivated me to contribute even more in the final stages. The experience reinforced the importance of clear documentation, mutual respect, and shared responsibility.

Closing Statement

Overall, this capstone experience sharpened both my technical competencies and my ability to collaborate in a high-stakes, team-oriented environment. It emphasized the importance of adaptability, clarity in communication, and designing infrastructure with purpose. I'm leaving this course more prepared for a professional role in cybersecurity and network design.

References

- Meyers, M. (2022). *CompTIA Network+ Certification Passport* (7th ed.). McGraw-Hill Education.
- U.S. Cybersecurity and Infrastructure Security Agency. (n.d.). *Securing network infrastructure devices*. <https://www.cisa.gov>
- Abbas, H., Emmanuel, N., Amjad, M. F., Yaqoob, T., Atiquzzaman, M., Iqbal, Z., Shafqat, N., Shahid, W. B., Tanveer, A., & Ashfaq, U. (2023). Security assessment and evaluation of VPNs: A comprehensive survey. *ACM Computing Surveys*, 55(13s), 1–47.
<https://doi.org/10.1145/3579162>